# Lab Exercise 16- Creating a ConfigMap from a Folder

In this lab exercise, you will learn how to create a ConfigMap in Kubernetes from a folder containing multiple files. This allows you to manage multiple configuration files as a single ConfigMap.

**Step 1: Set Up Kubernetes Cluster**

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

**Step 2: Prepare the Folder with Multiple Files**

Create a directory named configmap-folder:

```
mkdir configmap-folder
cd configmap-folder/
```

```
mohdanas@Mohds-MacBook-Air Desktop % mkdir configmap-folder
cd configmap-folder
```

Create multiple configuration files inside the directory:

**config1.txt** with content:

```
key1=value1
```

**config2.txt** with content:

```
key2=value2
```

**config3.txt** with content:

```
key3=value3
```

```
mohdanas@Mohds-MacBook-Air configmap-folder % echo "key1=value1" > config1.txt
echo "key2=value2" > config2.txt
echo "key3=value3" > config3.txt
```

Verify the directory structure:

```
ls configmap-folder/
cd ..
```

```
mohdanas@Mohds-MacBook-Air configmap-folder % ls
config1.txt     config2.txt     config3.txt
```

You should see the files config1.txt, config2.txt, and config3.txt.

### Step 3: Create a ConfigMap from the Folder

Create a ConfigMap named folder-configmap from the folder:

```
kubectl create configmap folder-configmap --from-file=configmap-folder/
```

```
mohdanas@Mohds-MacBook-Air configmap-folder % kubectl create configmap folder-configmap \
--from-file=. \
-n myns
configmap/folder-configmap created
```

Verify the ConfigMap:

```
kubectl get configmap folder-configmap -o yaml
```

You should see the folder-configmap with the contents of all the files in the configmap-folder.

```
[mohdanas@Mohds-MacBook-Air configmap-folder % kubectl get configmap folder-configmap -n myns -o yaml  ]
 apiVersion: v1
 data:
   config1.txt: |
     key1=value1
   config2.txt: |
     key2=value2
   config3.txt: |
     key3=value3
 kind: ConfigMap
 metadata:
   creationTimestamp: "2026-02-28T17:36:36Z"
   name: folder-configmap
   namespace: myns
   resourceVersion: "11208"
   uid: c9b3da38-0560-4eec-a8de-0a54cb3b7bcd
```

### Step 4: Use the ConfigMap in a Pod

Create a file named **pod-folder-configmap.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
        name: folder-configmap
```

In this manifest:

- The Pod mounts the folder-configmap at /etc/config.
- The busybox container concatenates and prints the contents of the files.

Apply the manifest to create the Pod:

```
kubectl apply -f pod-folder-configmap.yaml
```

```
mohdanas@Mohds-MacBook-Air configmap-folder % kubectl apply -f pod-folder-configmap.yaml
pod/folder-configmap-pod created
```

Check the status of the Pod:

```
kubectl get pods
```

```
mohdanas@Mohds-MacBook-Air configmap-folder % kubectl get pods -n myns
NAME                    READY   STATUS    RESTARTS   AGE
folder-configmap-pod    1/1     Running   0          42s
```

View the Pod logs to see the content of the mounted files:

```
kubectl exec -it configmap-pod -- printenv
```

```
mohdanas@Mohds-MacBook-Air configmap-folder % kubectl exec -it folder-configmap-pod -n myns -- /bin/sh

# cd /etc/config
# ls
config1.txt  config2.txt  config3.txt
# cat config1.txt
key1=value1
#
```

You should see the contents of config1.txt, config2.txt, and config3.txt printed.

**Step 5: Clean Up**

After completing the exercise, clean up the resources created:

```
kubectl delete pod folder-configmap-pod
kubectl delete configmap folder-configmap
rm -r configmap-folder
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl delete pod folder-configmap-pod -n myns
kubectl delete configmap folder-configmap -n myns
rm -r configmap-folder
pod "folder-configmap-pod" deleted
configmap "folder-configmap" deleted
rm: configmap-folder: No such file or directory
```